An investigation into Lightweight Hand Pose estimation for Edge HCI

Brandon MacGillivray
School of Electronics, Electrical Engineering and Computer Science
Queen's University Belfast, Belfast, United Kingdom
Email: bmacgillivray01@qub.ac.uk

***Abstract*—**Hand pose estimation is increasingly important for touchless human-computer interaction in settings such as AR/VR, public displays, and clinical or assistive environments, but deploying it on edge devices remains difficult because systems must balance prediction quality, latency, and robustness under limited compute and imperfect training data. This paper investigates lightweight 2D RGB hand pose estimation for that setting using a DRHand-inspired dual-branch architecture. The study examines whether a reduced 10-keypoint representation can retain sufficient utility for interaction-oriented use, whether dual-branch fusion improves prediction relative to the individual heatmap and coordinate branches, and whether pseudo-labeled real-image data can improve generalisation beyond synthetic-only training. The implemented pipeline is evaluated on RHD and the Hand Keypoint Dataset 26K (HK26K), together with transfer-learning runs between the two domains, comparison against a MediaPipe baseline, deployment-oriented benchmarking on Jetson Orin Nano, and qualitative analysis on selected benchmark and real-image examples. The results show that the reduced 10-keypoint model preserves most practically useful predictive performance while offering a modest runtime advantage, suggesting that full 21-keypoint prediction is not always necessary for lightweight edge-HCI settings. Fusion provides the strongest overall balance of results, although diagnostic analysis shows that the current heuristic remains suboptimal. Transfer learning substantially improves cross-domain performance when used for target-domain adaptation, and the resulting DRHand-inspired variant achieves practical embedded runtime.

***Keywords*—**hand pose estimation; human-computer interaction; edge devices; lightweight deep learning; transfer learning; embedded vision

I INTRODUCTION

Hand pose estimation is a useful interface modality for touchless HCI, particularly in contexts where accessibility, hygiene, or hands-free interaction are important. Prior HCI research describes hand-based interaction as natural and expressive, and demonstrates its use for pointing, manipulation, and contact-free control in settings such as AR/VR, large displays, and clinical environments [1–3]. In these scenarios, avoiding physical contact is advantageous when conventional input devices are inconvenient, undesirable, or impractical. Unlike simple gesture classification, landmark-based hand pose estimation provides a structured representation that supports richer and more flexible interaction, making it a suitable target for practical touchless HCI research.

A significant technical challenge is achieving useful hand-pose performance under edge-device constraints. Real-world deployment requires balancing prediction quality with latency, throughput, and computational cost on constrained hardware rather than desktop GPUs [4, 6, 9]. A highly accurate model is insufficient for HCI applications if it cannot respond quickly enough in embedded settings. Thus, this paper focuses on lightweight RGB hand-pose estimation for edge deployment.

Existing lightweight hand-pose methods leave open important design questions regarding keypoint representation, the value of dual-branch fusion, and generalisation under limited training data. MediaPipe Hands demonstrates efficient on-device tracking, while DRHand combines heatmap and coordinate regression in a lightweight dual-branch framework and achieves competitive performance [4, 5]. In many hand-pose systems, the hand is represented as a fixed set of landmark keypoints, typically a standard 21-keypoint skeleton covering the wrist and finger joints. Although these approaches establish the feasibility of real-time hand tracking, they do not fully resolve whether full 21-keypoint prediction is necessary for practical use, whether fusion provides sufficient benefit to justify its added complexity, or how robustness can be improved when training data are limited. This motivates a targeted study of reduced keypoint prediction, fusion utility, and cross-model knowledge transfer.

This paper investigates these questions through a DRHand-inspired implementation and a deployment-oriented evaluation. The study compares the full 21-keypoint and reduced 10-keypoint layouts shown in Figure 1, evaluates fusion against the individual branches, and analyses transfer between RHD and HK26K, where the latter uses MediaPipe-generated pseudo-labels. It also includes comparison against a MediaPipe baseline and runtime benchmarking on NVIDIA Jetson Orin Nano hardware. These experiments are designed to assess both prediction quality and practical deployment trade-offs. The contribution of the paper is therefore an empirical study examining current assumptions and adaptations to dataset constraints.

II LITERATURE REVIEW

The literature does not converge on a single architectural strategy for efficient pose estimation. Existing approaches include CNN-based systems such as MediaPipe Hands [5], attention-based models [9], hybrid CNN-transformer designs such as EdgePose [14], and temporal GCN-based tracking methods [10]. This diversity suggests that no single dominant architecture has yet been established for efficient hand-pose estimation. It is therefore reasonable to evaluate a DRHand-inspired architecture rather than assume that the design space has already been settled.

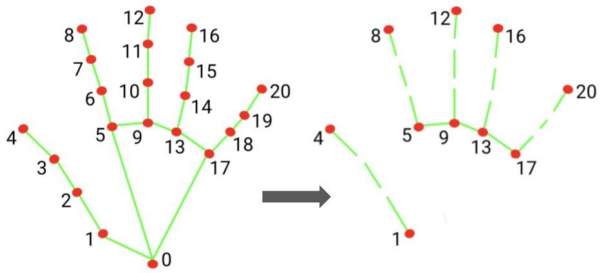

Figure 1: Keypoint layouts considered in this study. Left: full 21-keypoint hand skeleton. Right: reduced 10-keypoint skeleton using finger bases and fingertips.

Edge deployment imposes a speed–accuracy trade-off in pose estimation. Prior work such as SPLite Hand [12] explicitly addresses this trade-off, while other approaches assume access to desktop-level compute. This reflects a persistent mismatch in the literature between models optimised for benchmark accuracy and those that are practical for embedded hardware. Accordingly, runtime and deployment cost should be treated as core evaluation criteria.

Lightweight pose estimation methods typically achieve efficiency through a combination of architectural, computational, and pipeline-level simplifications. Reported strategies include the use of compact backbones and lightweight attention mechanisms [9], sparse or reduced-complexity computation [12], and detector-to-tracker pipelines with cropped regions of interest [5]. These approaches suggest that efficient performance is generally obtained by reducing unnecessary computation across multiple stages of the model rather than through a single isolated modification. This provides literature support for investigating whether simpler prediction targets can preserve useful hand-pose performance.

The realism and domain characteristics of training data are important determinants of cross-domain generalisation in hand-pose estimation. Prior work makes use of both synthetic datasets such as RHD and real datasets for hand tracking, with augmentation often introduced to improve robustness under more realistic conditions [10, 15, 16]. These data sources differ significantly in appearance, meaning that a model trained on one domain may not generalise reliably to another, particularly when synthetic supervision is used but deployment targets real camera input. Thus, dataset realism and cross-domain robustness remain important considerations in lightweight hand-pose estimation.

The literature leaves open whether full 21-keypoint hand representations are always necessary for lightweight edge-HCI use. Existing lightweight hand-pose systems such as MediaPipe Hands [5], Attention! [9], and SPLite Hand [12] primarily focus on reducing model complexity while still predicting a complete hand structure. This suggests that prior work has concentrated more on compressing the model than on questioning whether the full landmark target is required for practical HCI interaction. This gap directly motivates the present study's comparison between reduced 10-keypoint and standard 21-keypoint prediction.

The three main research questions:

1. Is a reduced 10-keypoint representation sufficient for lightweight edge-HCI use?
2. Does dual-branch fusion provide enough benefit to justify its added complexity?
3. Can pseudo-labeled real-image data improve generalisation beyond synthetic-only training?

III METHOD

A Network Structure

The proposed model uses a shared backbone followed by two prediction branches. The input is a resized RGB hand image, and most computation is shared before the network splits into heatmap and coordinate regression paths. Here, the term *branch* refers to a task-specific prediction subnetwork attached to the shared backbone, specifically the heatmap regression branch and the coordinate regression branch. Figures 2 and 3 show the overall DRHand-inspired network structure used in this study.

A.1 The Backbone Network

The backbone is designed to extract useful hand features. Formally, the input to the model is an RGB image resized to 256×256, denoted by $x \in \mathbb{R}^{3\times256\times256}$. The backbone then produces a compact feature representation,

$$F = B(x). \tag{1}$$

The backbone is built from depthwise-separable convolutions, downsampling blocks, and skip-connected upsampling stages.

A.2 The Heatmap Regression Branch

The heatmap branch predicts one spatial confidence map per keypoint. From the shared representation, the heatmap branch predicts K output heatmaps,

$$\hat{H} = R_h(F). \tag{2}$$

Here, K is the number of predicted keypoints, so $\hat{H} \in \mathbb{R}^{K\times64\times64}$ contains one 64×64 heatmap for each keypoint. For keypoint i, $\hat{H}_i(u,v)$ denotes the predicted confidence at spatial location (u,v) on the 64×64 heatmap grid, where u and v are the horizontal and vertical heatmap indices respectively. Let $p_i = (x_i, y_i)$ denote the ground-truth location of keypoint i, expressed in the same heatmap coordinate system. The target heatmap for keypoint i is then defined as a 2D Gaussian centered at the true keypoint location,

$$G_i(u,v) = \exp\left(-\frac{(u-x_i)^2 + (v-y_i)^2}{2\sigma^2}\right), \tag{3}$$

where $\sigma > 0$ is the standard deviation of the Gaussian, measured in heatmap-grid units, and controls the spread of the target peak. A larger σ produces a broader peak, while a smaller σ produces a sharper target concentrated around the true landmark.

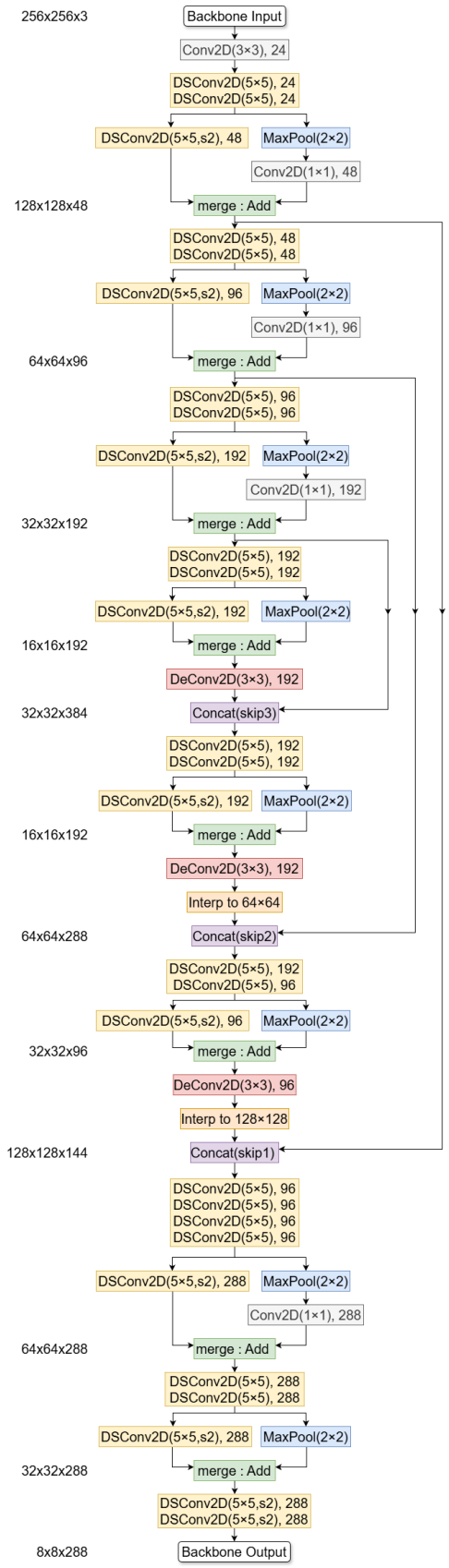

Figure 2: The backbone network structure.

The heatmap branch is supervised using mean-squared error between the predicted heatmaps and the target heatmaps:

$$\mathscr{L}_{hm} = \frac{1}{KHW} \sum_{i=1}^{K} \sum_{u,v} \left(\hat{H}_i(u,v) - G_i(u,v) \right)^2 \tag{4}$$

Here, H and W denote the heatmap height and width. This loss averages the squared difference over all keypoints and all spatial locations, encouraging each predicted heatmap to assign a high response near the true keypoint position and a low response elsewhere.

A.3 The Coordinate Regression Branch

The coordinate branch predicts landmark positions directly from intermediate features. Rather than producing a spatial heatmap for each keypoint, this branch regresses the landmark coordinates directly, taking as input the intermediate 64×64 feature map produced within the heatmap pathway. This allows the coordinate branch to reuse features that already encode spatial hand structure.

The coordinate branch predicts normalised landmark coordinates,

$$\hat{P}^{coord} = R_c(M), \qquad \hat{P}^{coord} \in \mathbb{R}^{K \times 2}, \tag{5}$$

where M denotes the intermediate 64×64 feature map passed from the heatmap branch to the coordinate regressor, $R_c(\cdot)$ denotes the coordinate-regression subnetwork, and K is the number of predicted keypoints. For keypoint i, the output $\hat{p}_i^{coord} = (\hat{x}_i, \hat{y}_i)$ represents the predicted normalised 2D coordinate, with both values expressed in the range $[0,1]$.

Let $p_i = (x_i, y_i)$ denote the corresponding ground-truth normalised coordinate for keypoint i. The coordinate branch is supervised using Wing loss,

$$\mathscr{L}_{coord} = \frac{1}{2K} \sum_{i=1}^{K} \sum_{j \in \{x,y\}} \phi\left(\hat{p}_{i,j}^{coord} - p_{i,j} \right), \tag{6}$$

where $\hat{p}_{i,j}^{coord}$ and $p_{i,j}$ are the predicted and ground-truth values of the j-th coordinate component, and

$$\phi(\delta) = \begin{cases} w \ln\left(1 + \frac{|\delta|}{\varepsilon}\right), & |\delta| < w, \\ |\delta| - C, & \text{otherwise}, \end{cases} \tag{7}$$

where

$$C = w - w \ln\left(1 + \frac{w}{\varepsilon}\right). \tag{8}$$

δ is the scalar prediction error for one coordinate value, w controls the range over which the loss behaves logarithmically, ε controls the curvature near zero, and C is a constant chosen so that the piecewise function is continuous at $|\delta| = w$.

Wing loss places greater emphasis on reducing small and medium localisation errors than standard L_1 or L_2 losses, while remaining less sensitive to very large errors.

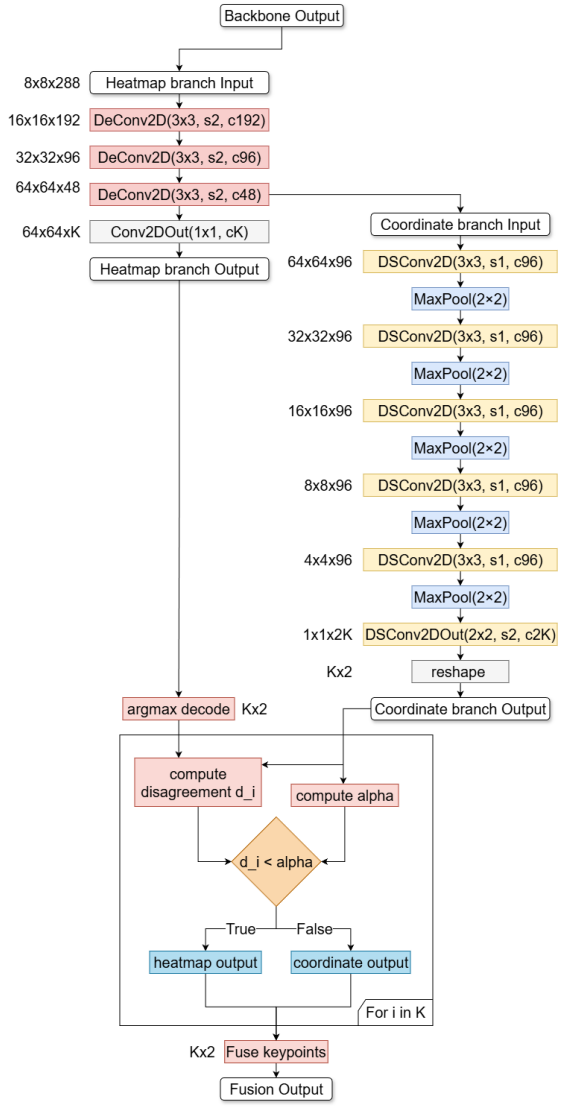

Figure 3: The heatmap, coordinate, and fusion components.

B Post Processing

The final pose can be produced by either branch alone or by a simple fusion rule. The implementation supports three inference modes: heatmap-only, coordinate-only, and fusion. In heatmap-only mode, landmark locations are obtained by decoding the peak of each predicted heatmap, while in coordinate-only mode the direct regression output is used as the final pose. In fusion mode, the two predictions are compared joint-by-joint and a simple heuristic selects which branch to trust for each landmark.

At inference time, let $\hat{H}_i$ denote the predicted heatmap for keypoint i, with spatial dimensions $H \times W$. The heatmap-based prediction is obtained by taking the spatial argmax,

$$(u_i^*, v_i^*) = \arg\max_{u,v} \hat{H}_i(u,v), \tag{9}$$

where (u_i^*, v_i^*) is the location of the peak response on the discrete heatmap grid. This peak is then converted into normalized image coordinates,

$$\hat{p}_i^{hm} = \left(\frac{u_i^*}{W-1}, \frac{v_i^*}{H-1} \right). \tag{10}$$

The coordinate branch directly predicts the normalized coordinate $\hat{p}_i^{coord}$ for the same joint, so both outputs lie in the same $[0,1]$ coordinate space and can be compared directly.

For each joint i, the disagreement between the two branches is measured by Euclidean distance,

$$d_i = \left\| \hat{p}_i^{hm} - \hat{p}_i^{coord} \right\|_2. \tag{11}$$

A sample-specific threshold α is then computed from the coordinate-branch output as the median predicted bone length,

$$\alpha = \text{median}_{(a,b)\in\mathscr{E}} \left\| \hat{p}_a^{coord} - \hat{p}_b^{coord} \right\|_2, \tag{12}$$

where $\mathscr{E}$ denotes the set of hand-skeleton edges. α acts as a scale-adaptive tolerance: for a larger predicted hand, a larger disagreement between the two branches may still be acceptable, whereas for a smaller predicted hand, the same disagreement would be more significant.

The fused output for joint i is defined as

$$\hat{p}_i^{fused} = \begin{cases} \hat{p}_i^{hm}, & d_i < \alpha, \\ \hat{p}_i^{coord}, & \text{otherwise.} \end{cases} \tag{13}$$

Thus, fusion is performed independently for each joint. When the two branches agree closely, the heatmap-decoded prediction is used; otherwise, the coordinate-regression prediction is retained.

IV EXPERIMENTS

A Datasets

The Rendered Handpose Dataset (RHD) [19] is used as the main synthetic benchmark in this study. It provides 41,258 training samples and 2,728 testing samples. Each sample contains a Blender-rendered 320×320 RGB hand image, together with 21 hand keypoints with image-space coordinates and visibility labels. The provided training/evaluation split is used.

The second dataset, HK26K (Kaggle) [20], is the more realistic image domain for transfer learning. It contains 18,776 training images and 7,992 validation images. The images are drawn from hand-photo and gesture datasets such as 11K Hands [21], 2000 Hand Gestures [22], and the Gesture Recognition Dataset [23], and include both hand-only views and realistic photographed scenes. The annotations are generated using MediaPipe and act as pseudo-labels. The provided training/evaluation split is used.

A pre-defined subset of an 8-image real-hand evaluation set (RH8) is also used for evaluation. This set contains 8 original real RGB hand images showing basic single-hand gestures on a white background, with annotations initialized from MediaPipe predictions and then manually corrected.

B Implementation Details

The proposed network was implemented in PyTorch and trained on an NVIDIA Tesla A100 80GB GPU, hosted on

Kelvin2 HPC, using the Adam optimizer. Two output configurations were used in this study: a standard 21-keypoint model and a reduced 10-keypoint variant. In both cases, the same overall architecture was retained, with only the output dimensionality of the prediction heads changed. Training followed a two-stage procedure. In stage 1, only the backbone and heatmap branch were optimized with learning rate 10^{-3} and batch size 64. In stage 2, the coordinate branch was added and the model was further optimized with learning rate 10^{-4} and an effective batch size of 256, implemented as batch size 64 with gradient accumulation. Runtime benchmarking for deployment-oriented evaluation was performed separately on Jetson Orin Nano 8GB, using its embedded Ampere GPU.

C Differences in this implementation and DRHand

This repository is a DRHand-inspired implementation rather than a strict reproduction of the original paper. The original work was implemented in TensorFlow and trained on YouTube2D Hands and GANerated Hands, whereas the present study is implemented in PyTorch and focuses on RHD together with pseudo-labeled HK26K data. Some optimisation details were not fully specified in the original paper, so several training values had to be reconstructed in this implementation. The reported results should therefore be interpreted as evidence about this reconstructed DRHand-style pipeline rather than as a direct numerical replication of the original study.

D Evaluation Metrics

Three main pose-estimation metrics are used to evaluate prediction quality. The learned models are evaluated using sum of squared error (SSE), end-point error (EPE), and probability of correct keypoint (PCK). Let $p_{si}, \hat{p}_{si} \in [0,1]^2$ denote the ground-truth and predicted normalized 2D coordinates of keypoint i in sample s, and let $v_{si} \in \{0,1\}$ denote a visibility mask, where $v_{si} = 1$ if the evaluated keypoint is visible and $v_{si} = 0$ otherwise. Invisible keypoints therefore do not contribute to the reported error values, and all distances are measured as Euclidean distances in normalized coordinate space.

The normalised SSE is defined as

$$\mathrm{SSE} = \frac{1}{N} \sum_{s=1}^{N} \sum_{i=1}^{K} v_{si} \, \|\hat{p}_{si} - p_{si}\|_2^2 , \tag{14}$$

where N is the number of samples and K is the number of evaluated keypoints. This metric sums the squared localization error over all visible evaluated keypoints in each sample and then averages that total over the dataset. Lower SSE therefore indicates lower overall pose error.

End-point error is measured in a root-relative normalized form. EPE is computed after subtracting a designated root joint from both the prediction and the ground truth, removing global translation and measuring how well the relative hand structure is recovered. Let r denote the root keypoint used for the current evaluation layout. In the full layout, keypoint 0 is used when available, while in reduced or shared 10-keypoint evaluation, keypoint 1 is used when keypoint 0 is absent, as shown in Figure 1.

The normalized root-relative EPE is defined as

$$\mathrm{EPE} = \frac{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si} \, \|(\hat{p}_{si} - \hat{p}_{sr}) - (p_{si} - p_{sr})\|_2}{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si}}, \tag{15}$$

where $\hat{p}_{sr}$ and p_{sr} are the predicted and ground-truth coordinates of the root keypoint in sample s. This metric reports the mean visible-joint distance after root alignment, so lower EPE indicates better recovery of the relative pose configuration. EPE is reported only when a supported root keypoint is present in the evaluated layout.

PCK measures the proportion of visible keypoints localized within a fixed error threshold. In this study, a keypoint is counted as correct when its normalized distance from the ground truth is at most 0.2, following the original DRHand evaluation protocol. Higher PCK therefore indicates a larger fraction of acceptable landmark predictions under the same tolerance.

$$\mathrm{PCK}_{\sigma} = \frac{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si} \, \mathbf{1}\left(\|\hat{p}_{si} - p_{si}\|_2 \le \sigma\right)}{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si}}, \qquad \sigma = 0.2, \tag{16}$$

where $\mathbf{1}(\cdot)$ is the indicator function, which returns 1 when the condition is true and 0 otherwise. Thus, PCK is the fraction of visible evaluated keypoints whose localization error falls within the accepted threshold. In this evaluation section, σ denotes the PCK threshold.

Runtime is reported separately from pose accuracy. Deployment-oriented evaluation includes prediction time in milliseconds per image and throughput in images per second. For the Jetson measurements, both forward-only and end-to-end timings are reported to distinguish model computation from surrounding pipeline overhead.

Additional metrics are used for baseline and fusion-specific analysis. For the MediaPipe baseline, detection success rate is also tracked because the pipeline may fail before landmark prediction. For fusion analysis, the study reports how often the heatmap branch is selected, the degree of disagreement between branch predictions, the typical size of the adaptive fusion threshold, and how often the heuristic agrees with an oracle choice of the lower-error branch computed using ground truth.

Let $m_{si} \in \{0,1\}$ denote the branch-selection decision for keypoint i in sample s, where $m_{si} = 1$ if fusion selects the heatmap prediction and $m_{si} = 0$ if fusion selects the coordinate-regression prediction. The heatmap-selection and coordinate-selection rates are then

$$\mathrm{Sel}^{hm} = \frac{1}{NK} \sum_{s=1}^{N} \sum_{i=1}^{K} m_{si}, \qquad \mathrm{Sel}^{coord} = \frac{1}{NK} \sum_{s=1}^{N} \sum_{i=1}^{K} (1 - m_{si}). \tag{17}$$

These rates describe how often each branch contributes the final fused output across all evaluated joints.

An oracle branch label is defined for each visible evaluated keypoint by comparing the two branch errors directly against ground truth:

$$o_{si} = \mathbf{1}\left(\left\| \hat{p}_{si}^{hm} - p_{si} \right\|_2 \le \left\| \hat{p}_{si}^{coord} - p_{si} \right\|_2 \right). \tag{18}$$

TABLE I: Main quantitative comparison between the 21-keypoint baseline and the reduced 10-keypoint model on RHD. Lower SSE and EPE are better; higher PCK@0.2 is better.

Model	Keypoint protocol	SSE ↓	EPE ↓	PCK@0.2 ↑
Baseline-21	21-keypoint	0.3098 ± 0.0236	0.0779 ± 0.0054	$\mathbf{0.9112 \pm 0.0071}$
Baseline-21	shared 10-keypoint	$\mathbf{0.1511 \pm 0.0120}$	0.0764 ± 0.0039	0.9101 ± 0.0071
Reduced-10	10-keypoint	0.1574 ± 0.0226	$\mathbf{0.0706 \pm 0.0054}$	0.9099 ± 0.0046

The oracle-match rate is then

$$\text{OracleMatch} = \frac{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si} \, \mathbf{1}(m_{si} = o_{si})}{\sum_{s=1}^{N} \sum_{i=1}^{K} v_{si}}. \tag{19}$$

This metric reports how often the fusion heuristic selected the lower-error branch on visible evaluated keypoints. A higher oracle-match rate therefore indicates that the fusion rule more often makes the locally correct decision. Each learned-model configuration was trained and evaluated in five independent runs using different random seeds. Reported values for learned models are therefore given as mean ± standard deviation across those five runs.

V RESULTS AND ANALYSIS

A *Experimental Results*

The learned-model configurations are as follows: the standard 21-keypoint baseline is referred to as Baseline-21, the reduced 10-keypoint model as Reduced-10, the single-domain transfer baselines as RHD-only and HK26K-only, and the fine-tuned transfer models as HK26K→RHD and RHD→HK26K.

A.1 Reduced Keypoint Performance

The reduced 10-keypoint model retains broadly comparable performance to the 21-keypoint baseline on the shared subset. In Table I, the shared 10-keypoint evaluation of Baseline-21 gives SSE 0.1511, EPE 0.0764, and PCK 0.9101, while the 10-keypoint Reduced-10 model gives SSE 0.1574, EPE 0.0706, and PCK 0.9099. These values are very close, indicating that reducing the landmark set does not substantially weaken performance on the subset of keypoints most relevant to the reduced representation. This supports the claim that a smaller landmark set can remain viable for lightweight edge-HCI use.

A.2 Runtime and Deployment Cost

The runtime advantage of the 10-keypoint model appears mainly in fusion mode. In Tables II and III, the forward-time gap between Baseline-21 and Reduced-10 is much larger in fusion mode (27.43 vs 24.59 ms/image) than in heatmap mode (24.28 vs 24.00 ms/image) or coordinate mode (24.08 vs 23.91 ms/image). The detailed breakdown further shows that the largest difference occurs in the forward stage, while the surrounding pipeline costs are comparatively similar. Since the single-branch modes differ only slightly, the main efficiency gain of the reduced model appears to come from lower fusion-stage overhead rather than from a large reduction in shared network cost. This suggests that the practical speed benefit of the 10-keypoint model is tied mainly to the post-processing stage, while the minor yet consistent speed gains observed in single-branch modes warrant further investigation.

TABLE II: Edge-device latency summary for the main models. Forward time measures the prediction stage only, whereas end-to-end time includes image read, preprocessing, host-to-device transfer, prediction, and JSON write.

Model	Mode	Forward ms/img ↓	End-to-end ms/img ↓	Images/s ↑
Baseline-21	Fused	27.43 ± 0.45	36.81 ± 1.33	27.20 ± 0.97
Baseline-21	Heatmap-only	24.28 ± 0.35	33.19 ± 0.56	30.13 ± 0.50
Baseline-21	Coordinate-only	24.08 ± 0.44	33.17 ± 0.86	30.16 ± 0.77
Reduced-10	Fused	24.59 ± 0.61	33.31 ± 0.71	30.03 ± 0.64
Reduced-10	Heatmap-only	24.00 ± 0.34	32.64 ± 0.46	30.64 ± 0.43
Reduced-10	Coordinate-only	$\mathbf{23.91 \pm 0.29}$	$\mathbf{32.50 \pm 0.34}$	$\mathbf{30.77 \pm 0.32}$

TABLE III: Detailed latency breakdown for the main fused models. These metrics are available from the benchmark pipeline and help localize where runtime is spent.

Model	Read ms	Prep ms	H2D ms	Forward ms	Write ms	E2E ms
Baseline-21 fusion	4.89 ± 0.64	2.80 ± 0.13	0.32 ± 0.02	27.43 ± 0.45	0.67 ± 0.09	36.81 ± 1.33
Reduced-10 fusion	$\mathbf{4.49 \pm 0.08}$	$\mathbf{2.74 \pm 0.05}$	$\mathbf{0.31 \pm 0.01}$	$\mathbf{24.59 \pm 0.61}$	$\mathbf{0.52 \pm 0.02}$	$\mathbf{33.31 \pm 0.71}$

A.3 Branch Behaviour

Fusion provides the strongest overall balance of branch behaviour. In Table IV, fusion achieves the lowest SSE and EPE for both Baseline-21 and Reduced-10, while maintaining PCK values close to the best single-branch result. For Baseline-21, heatmap-only attains the highest PCK but at the cost of substantially worse EPE, whereas for Reduced-10, coordinate-only provides only a marginal PCK gain over fusion while fusion retains better SSE and EPE. This indicates that the two branches contribute complementary strengths, with the heatmap branch improving threshold-based correctness and the coordinate branch preserving stronger geometric structure. Fusion therefore provides the most balanced overall operating point across these error criteria, supporting its use as the main operating mode of the system.

A.4 Transfer and Real-Image Adaptation

The experiments show that synthetic-only training does not generalize reliably to more realistic image domains, but pseudo-labeled real-image supervision provides a useful adaptation bridge. In Table V and Table VI, the RHD-trained model performs strongly on RHD but poorly on HK26K, while the HK26K-trained model shows the opposite pattern. After target-domain adaptation, RHD→HK26K and HK26K→RHD both improve substantially over their cross-domain starting points. This interpretation is further supported by the small real-image benchmark in Table VII, where the HK26K-oriented models, especially HK26K-only and RHD→HK26K, perform more

TABLE IV: Branch ablation for the dual-regression design. Fusion is compared against the heatmap-only and coordinate-only outputs using the same trained checkpoints. Each model is evaluated using its corresponding keypoint protocol.

Model	Mode	SSE ↓	EPE ↓	PCK@0.2 ↑
Baseline-21	Fused	$\mathbf{0.3098 \pm 0.0236}$	$\mathbf{0.0779 \pm 0.0054}$	0.9112 ± 0.0071
Baseline-21	Heatmap-only	0.3139 ± 0.0684	0.0989 ± 0.0335	$\mathbf{0.9374 \pm 0.0066}$
Baseline-21	Coordinate-only	0.3134 ± 0.0229	0.0788 ± 0.0062	0.9113 ± 0.0071
Reduced-10	Fused	$\mathbf{0.1574 \pm 0.0226}$	$\mathbf{0.0706 \pm 0.0054}$	0.9099 ± 0.0046
Reduced-10	Heatmap-only	0.2213 ± 0.0880	0.0991 ± 0.0288	0.9022 ± 0.0393
Reduced-10	Coordinate-only	0.1599 ± 0.0218	0.0715 ± 0.0070	$\mathbf{0.9102 \pm 0.0048}$

TABLE V: Transfer-learning results for the four training sequences evaluated on HK26K, using fusion-mode evaluation.

Training sequence	HK26K SSE	HK26K EPE	HK26K PCK
RHD-only	3.7488 ± 0.7985	0.3365 ± 0.0075	0.3719 ± 0.0434
HK26K-only	**0.0764 ± 0.0016**	**0.0612 ± 0.0049**	0.9834 ± 0.0007
HK26K → RHD	1.6426 ± 0.2293	0.2678 ± 0.0114	0.5320 ± 0.0164
RHD → HK26K	0.0980 ± 0.0064	0.0840 ± 0.0018	**0.9837 ± 0.0007**

TABLE VI: Transfer-learning results for the four training sequences evaluated on RHD, using fusion-mode evaluation.

Training sequence	RHD SSE	RHD EPE	RHD PCK
RHD-only	**0.3091 ± 0.0269**	**0.0764 ± 0.0030**	**0.9143 ± 0.0097**
HK26K-only	1.4194 ± 0.2619	0.1609 ± 0.0207	0.3397 ± 0.0718
HK26K → RHD	0.4504 ± 0.0162	0.1020 ± 0.0039	0.8190 ± 0.0114
RHD → HK26K	0.5699 ± 0.0592	0.1276 ± 0.0149	0.7690 ± 0.0221

strongly than the RHD-only baselines. These results indicate that the primary challenge is the domain gap between synthetic training data and more realistic image conditions. Although pseudo-labeled real-image data does not provide perfect supervision, it appears to move the model closer to the real-image domain than synthetic-only training can achieve on its own. Therefore, the transfer results are best interpreted as evidence that pseudo-labeled real-image supervision can act as a practical adaptation bridge beyond synthetic-only training.

A small real-image benchmark was also included. In addition to the main benchmark results on RHD and HK26K, the models were evaluated on RH8, which contains eight real images with annotations initialized from MediaPipe predictions and then manually corrected. These results provide a small but explicit real-image benchmark alongside the larger synthetic and pseudo-labeled datasets. A corresponding benchmark comparison against MediaPipe on RHD is reported in Table VIII.

A.5 Fusion Diagnostics

The current fusion heuristic is useful but clearly imperfect. In Table IX, the oracle-match rate is only about 0.48 for Baseline-21 and 0.53 for Reduced-10, even though fusion improves aggregate accuracy in the branch-ablation study. This indicates that the heuristic often fails to select the lower-error branch on a per-joint basis, but still provides overall benefit because some correct decisions occur on more influential prediction cases. Therefore, the current fusion rule is beneficial, but it also remains one of the clearest opportunities for future improvement.

TABLE VII: Real-image benchmark evaluation on RH8. The annotations were initialized from MediaPipe predictions and then manually corrected. MediaPipe is a single run. All learned-model results use fusion-mode evaluation.

Method	Keypoint protocol	SSE ↓	EPE ↓	PCK@0.2 ↑
Baseline-21	shared 10-keypoint	0.3345 ± 0.0221	0.2815 ± 0.0129	0.7075 ± 0.0143
Reduced-10	10-keypoint	0.4350 ± 0.1153	0.2752 ± 0.0307	0.6475 ± 0.0555
RHD-only	shared 10-keypoint	0.3595 ± 0.0475	0.2706 ± 0.0214	0.7050 ± 0.0338
HK26K-only	shared 10-keypoint	0.1215 ± 0.0434	0.1155 ± 0.0286	0.9200 ± 0.0647
HK26K→RHD	shared 10-keypoint	0.3644 ± 0.0360	0.2481 ± 0.0090	0.7075 ± 0.0301
RHD→HK26K	shared 10-keypoint	0.1209 ± 0.0236	0.1043 ± 0.0151	0.9425 ± 0.0301
MediaPipe	21-keypoint	0.0012	0.0018	**1.0000**
MediaPipe	shared 10-keypoint	**0.0005**	**0.0017**	**1.0000**

TABLE VIII: Comparison between the MediaPipe Hand Landmarker baseline and the learned DRHand variants on RHD. MediaPipe detection success on RHD was 0.8933; the learned-model evaluations do not use a comparable detection stage.

Method	Keypoint protocol	SSE	EPE	PCK@0.2
MediaPipe	21-keypoint	1.7049	0.0833	0.7707
MediaPipe	shared 10-keypoint	0.8599	0.1378	0.7635
Baseline-21 fusion	21-keypoint	0.3098 ± 0.0236	0.0779 ± 0.0054	**0.9112 ± 0.0071**
Baseline-21 fusion	shared 10-keypoint	**0.1511 ± 0.0120**	0.0764 ± 0.0039	0.9101 ± 0.0071
Reduced-10 fusion	10-keypoint	0.1574 ± 0.0226	**0.0706 ± 0.0054**	0.9099 ± 0.0046

TABLE IX: Fusion diagnostics for the dual-regression post-processing rule. The oracle-match rate measures how often the final fusion choice matches the lower-error branch for visible joints. HM sel. reports the heatmap-selection rate, and the coordinate-selection rate is its complement.

Model	Keypoint protocol	HM sel.	Disagr. mean	Oracle match
Baseline-21	21-keypoint	0.426 ± 0.025	0.163 ± 0.022	0.481 ± 0.012
Baseline-21	shared 10-keypoint	0.407 ± 0.024	0.162 ± 0.018	0.474 ± 0.011
Reduced-10	10-keypoint	0.501 ± 0.053	0.171 ± 0.018	**0.529 ± 0.037**

A.6 Hyperparameter Sensitivity

The baseline hyperparameter setting is a compromise rather than an obviously dominant configuration. In Table X, several alternative settings improve one metric while worsening another; for example, `E1` improves EPE and PCK relative to Baseline-21, whereas `W2` achieves the lowest SSE, but no single configuration is uniformly best across all reported metrics. The hyperparameter study indicates that not all settings have equal impact. End-to-end runtime changes are small across all variants, suggesting that these hyperparameters have negligible effect on deployment cost, while accuracy is more sensitive to certain parameters. The Wing-loss parameter ε appears particularly influential, with `E1` improving EPE and PCK relative to the baseline, and `E2` noticeably worsening EPE. The heatmap width σ and the Wing-loss parameter w also affect the accuracy trade-off, though in a more mixed manner, with some variants improving SSE, EPE, or PCK at the expense of others. In comparison, the loss-weight variants produce smaller or less consistently beneficial changes, aside from `L1`, which degrades SSE. Overall, the baseline should be regarded as a practical compromise, with ε emerging as the most influential of the tested hyperparameters. As a result, the selected baseline represents a reasonable compromise rather than a clearly dominated operating point.

B Qualitative Results

B.1 Real-Image Behaviour

The real-image examples favour the HK26K-oriented models. In the two RH8 figures, HK26K-only and especially RHD→HK26K place the red landmarks closer to the green annotations than Baseline-21 and Reduced-10, particularly around the inner finger cluster. This suggests that the real-image examples are visually closer to the HK26K domain than to the synthetic RHD domain. The qualitative results therefore support the transfer findings on real-image adaptation.

TABLE X: Hyperparameter ablation relative to the Baseline-21 configuration, using 21-keypoint fused accuracy on RHD and fused end-to-end benchmark latency. `L1`/`L2` modify the relative heatmap and coordinate loss weights, `S1`/`S2` modify the heatmap Gaussian width σ, `W1`/`W2` modify the Wing-loss parameter w, and `E1`/`E2` modify the Wing-loss parameter ε.

Variant	Change vs. Baseline-21	SSE ↓	EPE ↓	PCK@0.2 ↑	E2E ms ↓
B0	baseline	0.3098 ± 0.0236	0.0779 ± 0.0054	0.9112 ± 0.0071	36.81 ± 1.33
L1	$\lambda_{hm} = 1.5, \lambda_{coord} = 0.5$	0.3583 ± 0.0534	0.0725 ± 0.0096	0.9081 ± 0.0092	35.77 ± 0.45
L2	$\lambda_{hm} = 0.5, \lambda_{coord} = 1.5$	0.3073 ± 0.0260	0.0812 ± 0.0022	0.9115 ± 0.0095	35.75 ± 0.49
S1	$\sigma = 1.5$	0.3252 ± 0.0381	0.0707 ± 0.0055	0.9116 ± 0.0086	$\mathbf{35.72} \pm \mathbf{0.25}$
S2	$\sigma = 3.0$	0.3137 ± 0.0488	0.0833 ± 0.0058	$\mathbf{0.9177} \pm \mathbf{0.0081}$	36.48 ± 0.53
W1	$w = 5.0$	0.3213 ± 0.0384	0.0737 ± 0.0058	0.9150 ± 0.0099	36.21 ± 0.39
W2	$w = 15.0$	$\mathbf{0.2979} \pm \mathbf{0.0441}$	0.0750 ± 0.0031	0.9140 ± 0.0124	35.91 ± 0.43
E1	$\varepsilon = 1.0$	0.2994 ± 0.0591	$\mathbf{0.0695} \pm \mathbf{0.0011}$	0.9175 ± 0.0113	35.84 ± 0.23
E2	$\varepsilon = 4.0$	0.3106 ± 0.0106	0.0873 ± 0.0056	0.9110 ± 0.0067	36.13 ± 0.52

B.2 Cross-Dataset Failure and Adaptation

The model trained only on HK26K generalizes poorly when evaluated on the RHD dataset. In the RHD examples `00143` and `00354`, the HK26K-only model produces landmark groups that are visibly detached from the hand. This represents a clear visual failure under cross-dataset domain shift rather than a minor local error, reinforcing the conclusion that zero-shot transfer from HK26K to RHD is weak.

Target-domain adaptation improves the transferred HK26K model on RHD. In `00354`, HK26K→RHD is visibly better aligned to the hand than HK26K-only, even though it is not the strongest prediction in the figure. This indicates that fine-tuning reduces the most obvious cross-domain failures, although it does not eliminate them completely. This supports the interpretation that transfer is useful primarily as initialization for subsequent target-domain adaptation.

RHD→HK26K is the most consistently strong learned model across the selected figures. In Figures 4 and 5, RHD→HK26K is among the closest predictions to the ground truth on the two real-image examples, and in Figures 6 and 7 it remains stable on the two RHD examples, unlike HK26K-only. This visual pattern is also consistent with the quantitative transfer results in Table V and Table VI. This indicates that RHD pretraining followed by HK26K adaptation transfers more consistently across the displayed cases than HK26K-only training, and the selected figures therefore visually support the transfer asymmetry observed in the quantitative results.

B.3 MediaPipe Comparison Caveat

In Figures 4 and 5, MediaPipe appears visually strong, but the real-image comparison is not fully independent. MediaPipe aligns closely with the annotations in the two RH8 figures, although those annotations were initialized from MediaPipe predictions before manual correction. This introduces a structural advantage for MediaPipe on that small benchmark set, so the real-image comparisons should be treated as illustrative rather than definitive.

VI CONCLUSION AND FUTURE WORK

A reduced 10-keypoint representation is sufficient for the lightweight edge-HCI setting studied here. The reduced model remains close to the baseline in predictive quality while offering a runtime advantage. This indicates that the full 21-keypoint output is not strictly necessary for the interaction-focused scope of this study, and therefore the first research question is answered positively.

Dual-branch fusion is justified, but only as a marginal improvement with minimal time expense. Fusion provides the best overall balance across the reported results, although the diagnostic analysis shows that the current selection rule is not fully optimal. This indicates that while the two-branch design is useful, the present heuristic does not extract its full potential benefit, and therefore the second research question is answered as a qualified yes.

Pseudo-labeled real-image data improves generalisation beyond synthetic-only training, especially when used for target-domain adaptation. The transfer results show that zero-shot cross-domain generalisation is weak, but fine-tuning on the target domain substantially improves performance. This indicates that pseudo-labeled real-image supervision is beneficial primarily as an adaptation mechanism rather than as a complete replacement for target-domain training, and therefore the third research question is answered positively, with this limitation.

Future work should redesign the architecture specifically for reduced 10-keypoint prediction. The current results show that the reduced target remains effective, but the model still follows essentially the same DRHand-inspired design as the 21-keypoint version. This suggests that a purpose-built reduced architecture may enable further reductions in model size and latency. The next step is therefore to test whether a smaller model can preserve the same practical accuracy in the 10-keypoint setting.

Real-image evaluation should be broadened beyond the current pilot set. The present study includes only a small manually annotated real-image dataset used for supplementary quantitative and qualitative comparison. This indicates that a larger independently annotated real-image benchmark is needed to assess practical robustness more convincingly, and future work should therefore prioritise broader real-world evaluation.

Fusion remains one of the clearest opportunities for improvement. The current diagnostics show that the heuristic often fails to select the lower-error branch, and the original DRHand formulation also leaves room for stronger branch combination [4]. This suggests that a more principled or learned fusion strategy could improve both robustness and accuracy, and future work should therefore investigate improved fusion design.

Future work should introduce an explicit hand-detection stage to standardise input before pose estimation. The current pipeline assumes a prepared hand region, whereas practical systems such as MediaPipe first localise the hand and then estimate landmarks from a normalized crop. A detection stage could produce more consistent hand scale, position, and framing, aligning the input distribution more closely with training data and reducing reliance on aggressive augmentation. Future work should therefore evaluate whether detector-based normalisation improves robustness and simplifies pose learning.

QUALITATIVE FIGURES

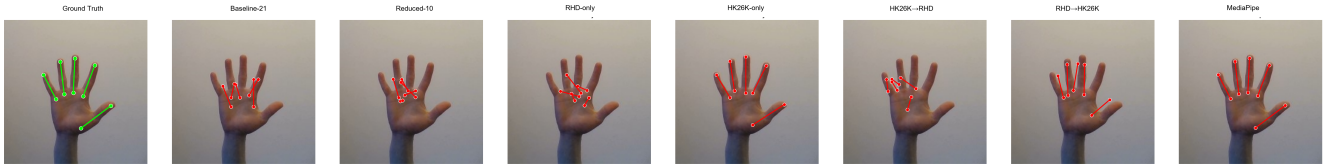

Figure 4: Qualitative comparison on RH8 example 1.

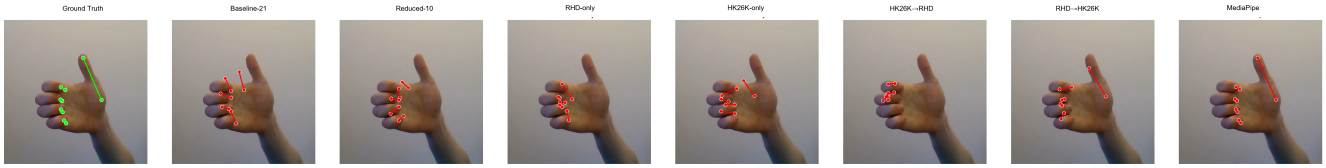

Figure 5: Qualitative comparison on RH8 example 6.

Figure 6: Qualitative comparison on RHD evaluation example 00143.

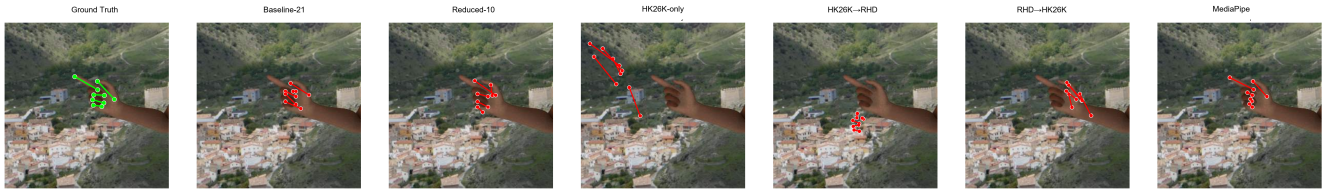

Figure 7: Qualitative comparison on RHD evaluation example 00354.

REFERENCES

[1] T. Vuletic, A. Duffy, L. Hay, C. McTeague, and G. Campbell, "Systematic literature review of hand gestures used in human-computer interaction interfaces," *Int. J. Human-Computer Studies*, vol. 129, pp. 47–65, 2019.

[2] J. P. Wachs, M. Kolsch, H. Stern, and Y. Edan, "Vision-based hand-gesture applications," *Communications of the ACM*, vol. 54, no. 2, pp. 60–71, 2011.

[3] Y. Li, J. Huang, F. Tian, H.-A. Wang, and G.-Z. Dai, "Gesture interaction in virtual reality," *Virtual Reality & Intelligent Hardware*, vol. 1, no. 1, pp. 84–112, 2019.

[4] D. Wei, S. An, X. Zhang, J. Tian, K. A. Tsintotas, A. Gasteratos, and H. Zhu, "Dual regression for efficient hand pose estimation," in *Proc. IEEE Int. Conf. Robotics and Automation*, 2022.

[5] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, and M. Grundmann, "MediaPipe hands: On-device real-time hand tracking," *arXiv preprint arXiv:2006.10214*, 2020.

[6] Y. Wang, B. Zhang, and C. Peng, "SRHandNet: Real-time 2D hand pose estimation with simultaneous region localization," *IEEE Trans. Image Processing*, vol. 29, pp. 2977–2986, 2019.

[7] Z.-H. Feng, J. Kittler, M. Awais, P. Huber, and X.-J. Wu, "Wing loss for robust facial landmark localisation with convolutional neural networks," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition*, 2018, pp. 2235–2245.

[8] S. An, X. Zhang, D. Wei, H. Zhu, J. Yang, and K. A. Tsintotas, "FastHand: Fast monocular hand pose estimation on embedded systems," *Journal of Systems Architecture*, vol. 122, p. 102361, 2022.

[9] N. Santavas, I. Kansizoglou, L. Bampis, E. Karakasis, and A. Gasteratos, "Attention! A lightweight 2D hand pose estimation approach," *IEEE Sensors Journal*, vol. 21, no. 10, pp. 11488–11501, 2021.

[10] Y. Yin, C. McCarthy, and D. Rezazadegan, "Real-time lightweight 3D hand-object pose estimation using temporal graph convolution networks," in *AI 2024: Advances in Artificial Intelligence*, LNCS 15443, 2024, pp. 243–255.

[11] A. Prakash, R. Tu, M. Chang, and S. Gupta, "3D hand pose estimation in everyday egocentric images," in *Proc. ECCV*, 2024.

[12] K. H. Yeh, T. W. Hsu, and M. Sun, "SPLite Hand: Sparsity-aware lightweight 3D hand pose estimation," in *Proc. AICCC 2025*, Tokyo, Japan, 2025, pp. 1–10.

[13] E. Williams-Linera and J. M. Ramirez-Cortes, "Stereo vision system based on the NVIDIA Jetson Nano for real-time evaluation of yoga poses," in *Proc. IEEE*, 2024, pp. 1–7.

[14] L. Zhang et al., "EdgePose: Real-time human pose estimation scheme for industrial scenes," *IEEE Access*, vol. 12, pp. 156702–156716, 2024.

[15] Y. Zhang et al., "A lightweight hand attitude estimation method based on GCN feature enhancement," *Electronics*, vol. 13, no. 10, p. 1952, 2024.

[16] Z. Zhang, S. Xie, M. Chen, and H. Zhu, "HandAugment: A simple data augmentation method for depth-based 3D hand pose estimation," *arXiv preprint arXiv:2001.00702*, 2020.

[17] I. B. Toft, J. P. B. Nielsen, N. K. Poulsen, and A. L. Laursen, "EfficientPose: Scalable single-person pose estimation," *arXiv preprint arXiv:2008.09062*, 2020.

[18] V. Bazarevsky, I. Grishchenko, K. Raveendran, T. Zhu, F. Zhang, and M. Grundmann, "BlazePose: On-device real-time body pose tracking," *arXiv preprint arXiv:2006.10204*, 2020.

[19] C. Zimmermann and T. Brox, "Learning to estimate 3D hand pose from single RGB images," in *Proc. IEEE Int. Conf. Computer Vision*, 2017.

[20] Rion Dsilva, "Hand Keypoint Dataset 26K," Kaggle dataset. Available: `https://www.kaggle.com/datasets/riondsilva21/hand-keypoint-dataset-26k`

[21] M. Afifi, "11K Hands: Gender recognition and biometric identification using a large dataset of hand images," *Multimedia Tools and Applications*, 2019.

[22] R. Giridhar, "Hand gestures dataset," Kaggle dataset, 2022. doi: `https://doi.org/10.34740/KAGGLE/DS/2206851`

[23] Sparsh, "Gesture recognition dataset," Kaggle dataset. Available: `https://www.kaggle.com/datasets/imsparsh/gesture-recognition/data`